

SDAR as a Per-Token Trust-Constraint Lagrangian

Improvement 104 – SDAR study

Setup

A *constrained Markov decision process* (CMDP) is a tuple $(\mathcal{S}, \mathcal{A}, P, R, \{C_i, \delta_i\}_{i=1}^k)$ where the agent maximizes the cumulative reward subject to expected-cost constraints $\mathbb{E}_\pi [\sum_t C_i(s_t, a_t)] \leq \delta_i$. The standard tool is the Lagrangian:

$$\mathcal{L}(\pi, \mu) = -\mathbb{E}_\pi \left[\sum_t R(s_t, a_t) \right] + \sum_{i=1}^k \mu_i \cdot \left(\mathbb{E}_\pi \left[\sum_t C_i \right] - \delta_i \right),$$

with $\mu_i \geq 0$. Convergence theory for CMDPs (Altman 1999; Borkar 2005; Tessler et al. 2018) establishes saddle-point conditions on (π^*, μ^*) under standard assumptions (convexity of the policy class in occupancy measures, Slater’s condition for strict feasibility, decaying step sizes for primal-dual updates).

In SDAR, the policy is parameterized by θ , the per-token quantities are Δ_t and g_t from the paper, and the GRPO term plays the role of $-\mathbb{E}_\pi [\sum_t R]$.

Mapping

Define the per-token trust constraint

$$\Phi(\theta) := \mathbb{E}_{\tau \sim \pi_\theta} \left[\frac{\sum_t m_t g_t \Delta_t}{\sum_t m_t} \right] = \mathbb{E}_{\tau \sim \pi_\theta} \left[\mathcal{L}_{\text{SDAR}}(\theta) \cdot \frac{1}{|\tau|} \right],$$

i.e. the expected SDAR auxiliary, summed (with stop-gradient on g_t). Consider the constrained problem

$$\max_{\theta} J_{\text{GRPO}}(\theta) \quad \text{subject to} \quad \Phi(\theta) \leq \delta, \tag{1}$$

for some slack $\delta \geq 0$.

Proposition 1 (Lagrangian correspondence). *Let $\mu \geq 0$ be a fixed Lagrange multiplier. The Lagrangian of (1) is*

$$\mathcal{L}_{\text{lagr}}(\theta, \mu) = -J_{\text{GRPO}}(\theta) + \mu \cdot (\Phi(\theta) - \delta) = \mathcal{L}_{\text{GRPO}}(\theta) + \mu \cdot \mathcal{L}_{\text{SDAR}}(\theta) - \mu\delta + (\text{constants}).$$

Up to the θ -independent constant $-\mu\delta$, this is exactly the SDAR combined objective with $\lambda_{\text{SDAR}} = \mu$. In particular, $\nabla_{\theta} \mathcal{L}_{\text{lagr}}(\theta, \mu) = \nabla_{\theta} [\mathcal{L}_{\text{GRPO}} + \mu \mathcal{L}_{\text{SDAR}}]$, so the SDAR training loop is the inner (θ) loop of a primal-dual scheme with the dual variable held constant.

Proof. Direct substitution. The masked-token average Agg used in $\mathcal{L}_{\text{SDAR}}$ commutes with the outer $\mathbb{E}_{\tau}$ when we move from per-token to per-trajectory aggregation (the mask weights are deterministic functions of the trajectory), so $\Phi(\theta) = \mathbb{E}_{\tau} [\mathcal{L}_{\text{SDAR}}]$ up to normalization conventions. The $-J_{\text{GRPO}} = \mathcal{L}_{\text{GRPO}}$ identification is the standard sign convention for the GRPO loss. $\square$

Theorem

Theorem 1 (CMDP convergence transfers to SDAR). *Suppose:*

1. *The policy parameterization π_θ admits a convex reformulation in occupancy measures (true for tabular policies; standard relaxation for over-parameterized neural policies, see Mei et al. 2020).*
2. *Slater’s condition holds for (1): there exists θ_0 with $\Phi(\theta_0) < \delta$.*
3. *Step sizes $\eta_\theta(k), \eta_\mu(k)$ for primal-dual updates satisfy $\sum_k \eta_\theta = \sum_k \eta_\mu = \infty$, $\sum_k \eta_\theta^2, \sum_k \eta_\mu^2 < \infty$, and $\eta_\mu/\eta_\theta \rightarrow 0$ (two-time-scale).*

Then the primal-dual updates

$$\begin{aligned}\theta_{k+1} &= \theta_k - \eta_\theta(k) \nabla_\theta [\mathcal{L}_{\text{GRPO}}(\theta_k) + \mu_k \mathcal{L}_{\text{SDAR}}(\theta_k)], \\ \mu_{k+1} &= \max\{0, \mu_k + \eta_\mu(k) (\Phi(\theta_k) - \delta)\},\end{aligned}$$

converge almost surely to a saddle point (θ^, μ^*) of $\mathcal{L}_{\text{lagr}}$, and θ^* solves (1).*

Sketch of translation. The above is the Borkar (2005) two-time-scale stochastic approximation result for constrained MDPs, instantiated for the specific cost Φ . The technical content is unchanged from the cited literature; the only translation step is verifying that the SDAR cost Φ satisfies the standard regularity (boundedness of $g_t \Delta_t$ on bounded-support gap distributions; Lipschitz continuity of $\nabla_\theta \mathcal{L}_{\text{SDAR}}$ via the gated REINFORCE form derived in concept 10). Both hold under the Assumption 1 of the bias-bound proof and standard smoothness assumptions on π_θ .

The converse direction – that the fixed- μ scheme used in the SDAR paper (where $\mu = \lambda_{\text{SDAR}}$ is set by hand and not updated) corresponds to a particular δ – follows from the strong duality of (1) under Slater: every dual variable μ corresponds to a unique constraint level $\delta(\mu)$ such that $\Phi(\theta^*) = \delta(\mu)$ at the dual optimum. $\square$

Discussion

What the connection buys.

- **Principled λ schedule.** Theorem 1 says the right way to choose λ_{SDAR} is by dual ascent on $\Phi - \delta$, with δ chosen on the validation set. This eliminates the need for a λ grid search (cf. improvement 103) and replaces it with a single tunable: the trust budget δ .
- **Convergence guarantee.** The fixed- λ scheme converges only to a stationary point of the joint loss. The primal-dual scheme converges to a saddle point of the constrained problem, which is provably feasible (i.e. the per-token trust constraint is satisfied at the limit). This is a stronger guarantee than the paper currently offers.
- **Multi-constraint generalization.** If we have multiple gating strategies (entropy, gap, soft-OR), each produces its own constraint $\Phi_i \leq \delta_i$. The Lagrangian becomes $\mathcal{L}_{\text{GRPO}} + \sum_i \mu_i \mathcal{L}_{\text{SDAR}}^{(i)}$, with each μ_i updated by its own dual-ascent step. This is the natural extension of SDAR to a portfolio of gating signals.
- **Connection to RLHF.** RLHF’s β KL-to-reference coefficient is also a fixed-multiplier instance of a CMDP (constrain KL to SFT $\leq \delta_{\text{KL}}$). The SDAR-as-CMDP framing thus puts SDAR and RLHF in the same formal category: both are KL-constrained policy improvements with primal-dual training.

Caveat. Two-time-scale primal-dual is empirically slower to converge than the fixed- μ scheme. So the practical question is whether the convergence guarantee (and the multi-constraint generalization) outweigh the wall-clock cost. We conjecture yes for production-scale runs and no for small-scale ablations.